

EXDA Gas Mixing Export

Project: VH2D-Project-Wk-22-2026

Responsible Researcher: PhD Student Javier I. Camacho

Total Records: 25

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MATLAB Verified: Yes

Verification File A: scripts/GasMixingVerificationFiles/H2_MFC_Fill_Calculator_v000.m

Verification File B: scripts/GasMixingVerificationFiles/AuxFcn_H2_MFC_FillCalculator_000.m

Shared Chamber Setup (applies to all runs unless noted otherwise):

L=0.900 m, W=0.900 m, H=0.900 m, Pipes+=VARIES m³, Hotwire+=0.000 m³, Welded-=0.000 m³, Bolts-=0.000 m³, Vcorr=VARIES L

Note: setup values vary across runs; detailed per-run setup table is included below.

Core Gas Mixing Results

Group	Run/Test	H2%	RH%	Pch(Pa)	Tch(K)	H2est(g)	H2corr(g)	H2inj(L)	MFC	t(s)	t(min)	Model	Basis	a
Prep-Wk22	Prep-Wk22-01	13.50	-	101325	292.15	8.276	Not corrected by model	98.41	20.00	301.3	5.0	-	-	-
Verif-Wk22	Verif-Wk22-01	13.50	-	101325	294.25	8.217	Not corrected by model	98.42	20.00	299.2	5.0	-	-	-
Verif-Wk22	Verif-Wk22-02	14.00	-	101325	293.15	9.384	Not corrected by model	111.97	20.00	334.8	5.6	-	-	-
Verif-Wk22	Verif-Wk22-03	16.00	-	101325	294.15	10.638	Not corrected by model	127.37	20.00	387.3	6.5	-	-	-
Verif-Wk22	Verif-Wk22-04	18.00	-	101325	291.65	12.070	Not corrected by model	143.29	20.00	439.5	7.3	-	-	-
VH2D-Wk22-01	VH2D-Wk22-01-01	16.00	-	101325	293.15	10.674	Not corrected by model	127.37	20.00	388.6	6.5	-	-	-
VH2D-Wk22-01	VH2D-Wk22-01-02	16.00	-	101325	292.05	10.714	Not corrected by model	127.37	20.00	390.1	6.5	-	-	-
VH2D-Wk22-01	VH2D-Wk22-01-03	18.00	-	101325	291.35	12.083	Not corrected by model	143.29	20.00	439.9	7.3	-	-	-

VH2D-Wk22-01	VH2D-Wk22-01-04	18.00	-	101325	290.85	12.103	Not corrected by model	143.29	20.00	440.7	7.3	-	-	-
VH2D-Wk22-01	VH2D-Wk22-01-05	19.00	-	101325	290.55	12.789	Not corrected by model	151.26	20.00	465.6	7.8	-	-	-
VH2D-Wk22-01	VH2D-Wk22-01-07	20.00	-	101325	290.56	13.462	Not corrected by model	159.22	20.00	490.1	8.2	-	-	-
VH2D-Wk22-02	VH2D-Wk22-02-01	18.00	-	101325	289.95	12.141	12.278	143.29	20.00	447.0	7.5	linear_	fractio	69
VH2D-Wk22-02	VH2D-Wk22-02-02	18.00	-	101325	291.15	12.091	12.278	143.29	20.00	447.0	7.5	linear_	fractio	69
VH2D-Wk22-02	VH2D-Wk22-02-03	20.00	-	101325	293.15	13.343	13.671	159.22	20.00	497.7	8.3	linear_	fractio	69
VH2D-Wk22-02	VH2D-Wk22-02-04	20.00	-	101325	292.15	13.388	13.671	159.22	20.00	497.7	8.3	linear_	fractio	69
VH2D-Wk22-03	VH2D-Wk22-03-01	18.00	-	101325	292.45	12.037	12.278	143.29	20.00	447.0	7.5	linear_	fractio	69
VH2D-Wk22-03	VH2D-Wk22-03-02	18.00	-	101325	292.25	12.045	12.278	143.29	20.00	447.0	7.5	linear_	fractio	69
VH2D-Wk22-03	VH2D-Wk22-03-03	20.00	-	101325	292.15	13.388	13.671	159.22	20.00	497.7	8.3	linear_	fractio	69
VH2D-Wk22-03	VH2D-Wk22-03-04	20.00	-	101325	291.65	13.411	13.671	159.22	20.00	497.7	8.3	linear_	fractio	69
VH2D-Wk22-03	VH2D-Wk22-03-05	20.00	-	101325	291.45	13.421	13.671	159.22	20.00	497.7	8.3	linear_	fractio	69
VH2D-Wk22-04	VH2D-Wk22-04-01	16.00	-	101325	291.75	10.725	10.885	127.37	20.00	396.3	6.6	linear_	fractio	69
VH2D-Wk22-04	VH2D-Wk22-04-02	16.00	-	101325	292.75	10.689	10.885	127.37	20.00	396.3	6.6	linear_	fractio	69
VH2D-Wk22-04	VH2D-Wk22-04-03	21.00	-	101325	293.05	14.015	14.367	167.18	20.00	523.1	8.7	linear_	fractio	69
VH2D-Wk22-04	VH2D-Wk22-04-04	21.00	-	101325	293.75	13.981	14.367	167.18	20.00	523.1	8.7	linear_	fractio	69
VH2D-Wk22-04	VH2D-Wk22-04-05	12.00	-	101325	295.05	7.954	8.100	95.53	10.00	589.8	9.8	linear_	fractio	69

Per-Run Chamber Setup (because values vary)

Group	Run/Test	L (m)	W (m)	H (m)	Pipes+ (m3)	Hotwire+ (m3)	Welded- (m3)	Bolts- (m3)	Vcorr(L)	Notes
Prep-Wk22	Prep-Wk22-01	0.900	0.900	0.900	0.000	0.000	0.000	0.000	729.00	-
Verif-Wk22	Verif-Wk22-01	0.900	0.900	0.900	0.000	0.000	0.000	0.000	729.00	-
Verif-Wk22	Verif-Wk22-02	0.900	0.900	0.900	0.071	0.000	0.000	0.000	799.80	-
Verif-Wk22	Verif-Wk22-03	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-
Verif-Wk22	Verif-Wk22-04	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-
VH2D-Wk22	VH2D-Wk22-01	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-
VH2D-Wk22	VH2D-Wk22-01	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-
VH2D-Wk22	VH2D-Wk22-01	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-
VH2D-Wk22	VH2D-Wk22-01	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-
VH2D-Wk22	VH2D-Wk22-01	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-
VH2D-Wk22	VH2D-Wk22-01	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-
VH2D-Wk22	VH2D-Wk22-02	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-
VH2D-Wk22	VH2D-Wk22-02	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-

VH2D-Wk22.	VH2D-Wk22-02-.	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-	
VH2D-Wk22.	VH2D-Wk22-02-.	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-	
VH2D-Wk22.	VH2D-Wk22-03-.	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-	
VH2D-Wk22.	VH2D-Wk22-03-.	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-	
VH2D-Wk22.	VH2D-Wk22-03-.	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-	
VH2D-Wk22.	VH2D-Wk22-03-.	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-	
VH2D-Wk22.	VH2D-Wk22-03-.	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-	
VH2D-Wk22.	VH2D-Wk22-04-.	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-	
VH2D-Wk22.	VH2D-Wk22-04-.	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-	
VH2D-Wk22.	VH2D-Wk22-04-.	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-	
VH2D-Wk22.	VH2D-Wk22-04-.	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-	
VH2D-Wk22.	VH2D-Wk22-04-.	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-	
VH2D-Wk22.	VH2D-Wk22-04-.	0.900	0.900	0.900	0.067	0.000	0.000	0.000	796.08	-	MFC set to 7.954 g (12
											vol.%) but stopped at vol.
											8% at location measured by
											H2CM. This was the initial
											plan, but the total hydrogen
											mass was injected.